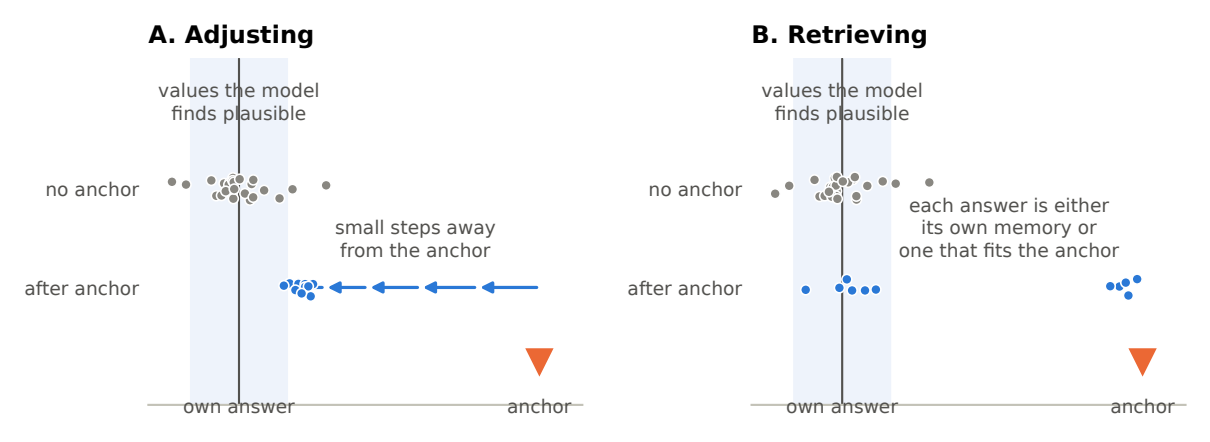
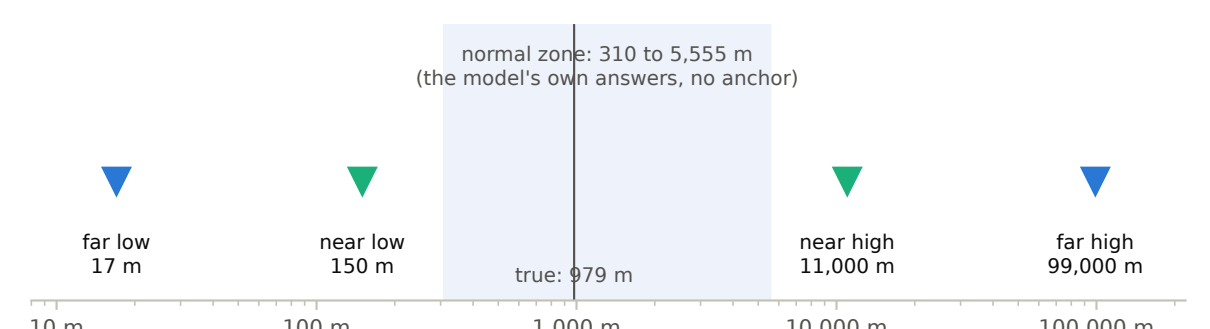
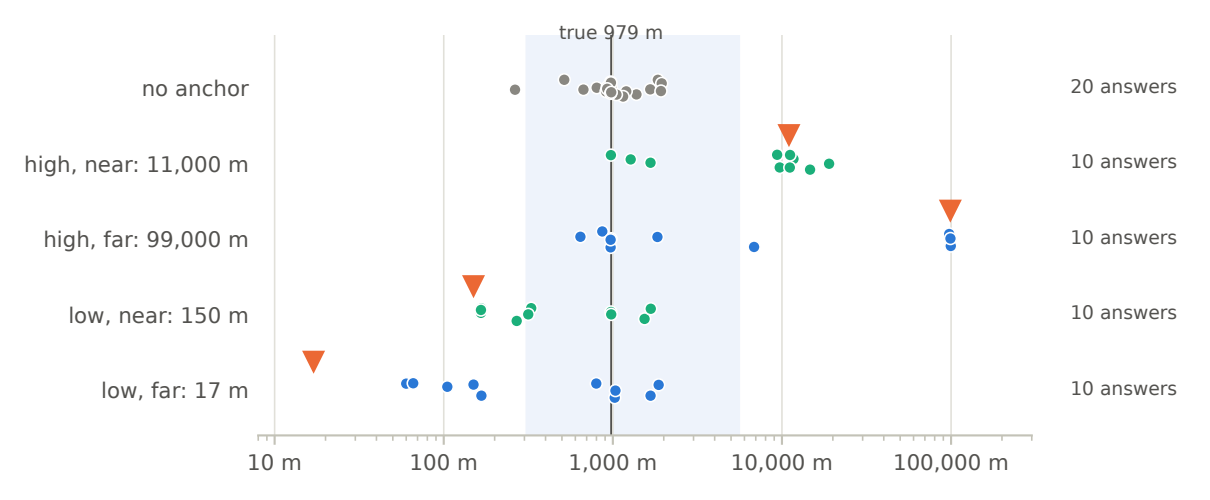
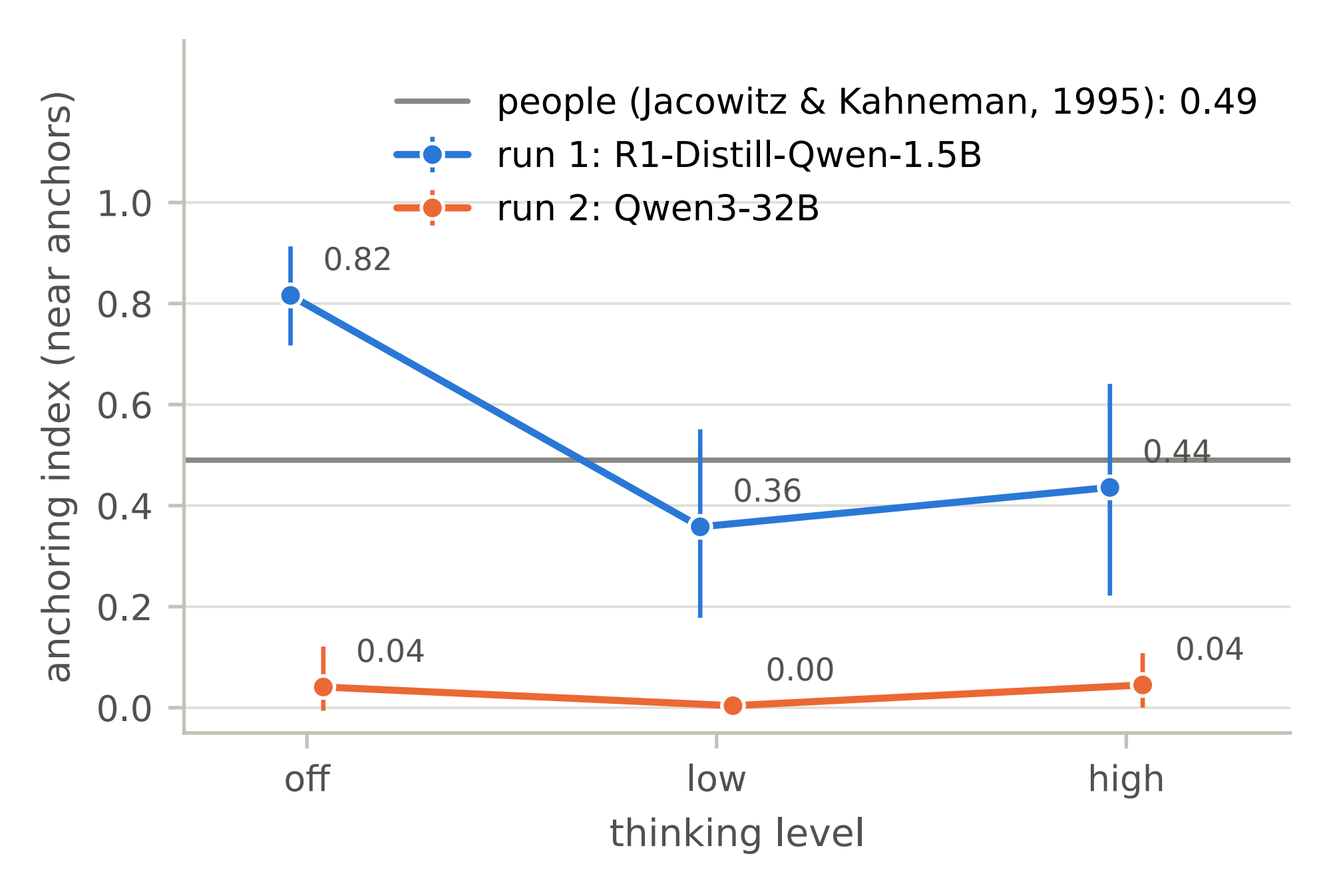
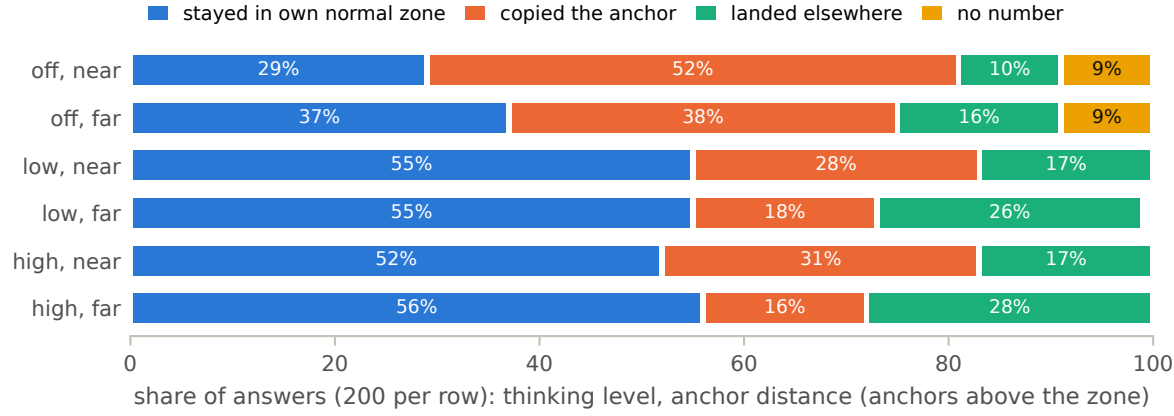
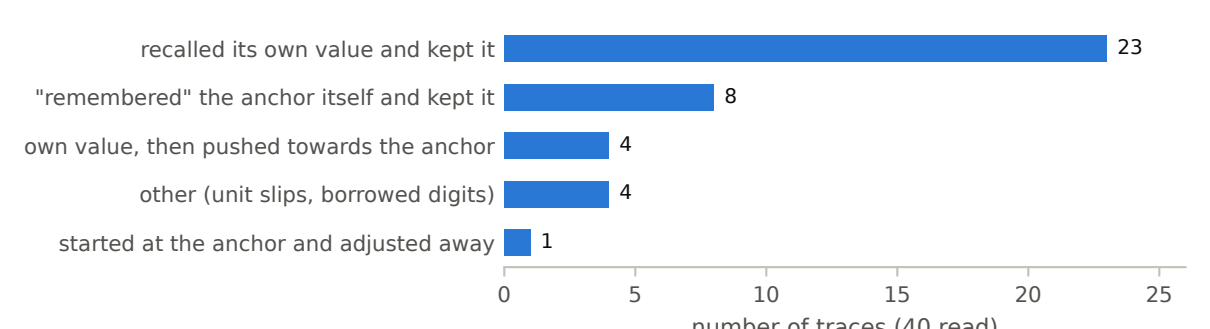

A. Adjusting
values the model finds plausible
no anchor
small steps away from the anchor
after anchor
own answer
anchor
B. Retrieving
values the model finds plausible
no anchor
each answer is either its own memory or one that fits the anchor
after anchor
own answer
anchor

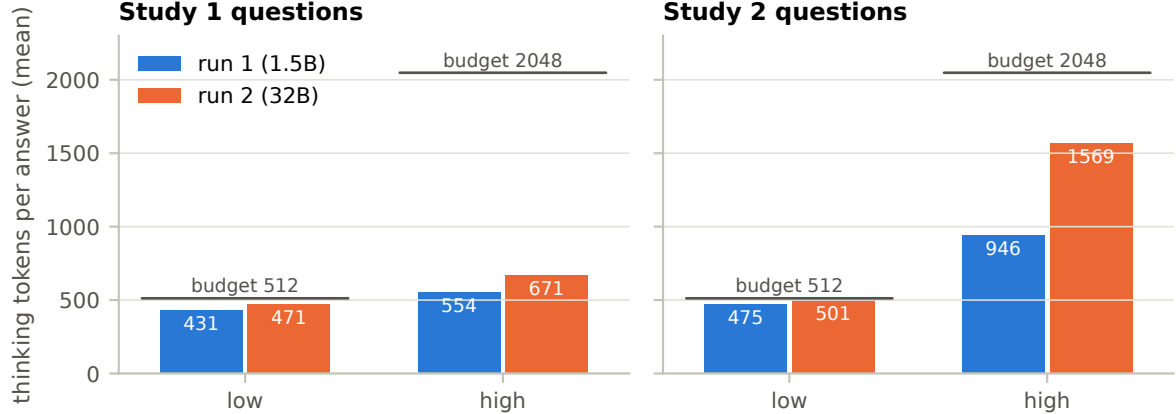

normal zone: 310 to 5,555 m
(the model's own answers, no anchor)
far low
17 m
near low
150 m
true: 979 m
near high
11,000 m
far high
99,000 m
10 m
100 m
1,000 m
10,000 m
100,000 m

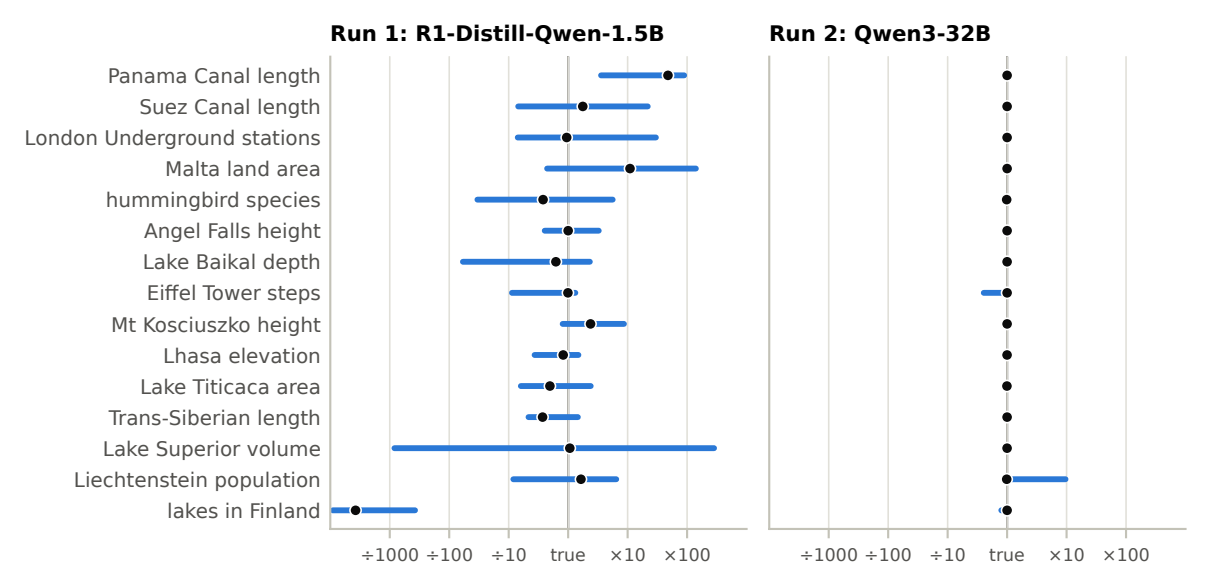

true 979 m
no anchor
high, near: 11,000 m
high, far: 99,000 m
low, near: 150 m
low, far: 17 m
20 answers
10 answers
10 answers
10 answers
10 answers
10 m
100 m
1,000 m
10,000 m
100,000 m


people (Jacowitz & Kahneman, 1995): 0.49
run 1: R1-Distill-Qwen-1.5B
run 2: Qwen3-32B
anchoring index (near anchors)
1.0
0.8
0.6
0.4
0.2
0.0
0.82
0.36
0.44
0.04
0.00
0.04
off
low
high
thinking level

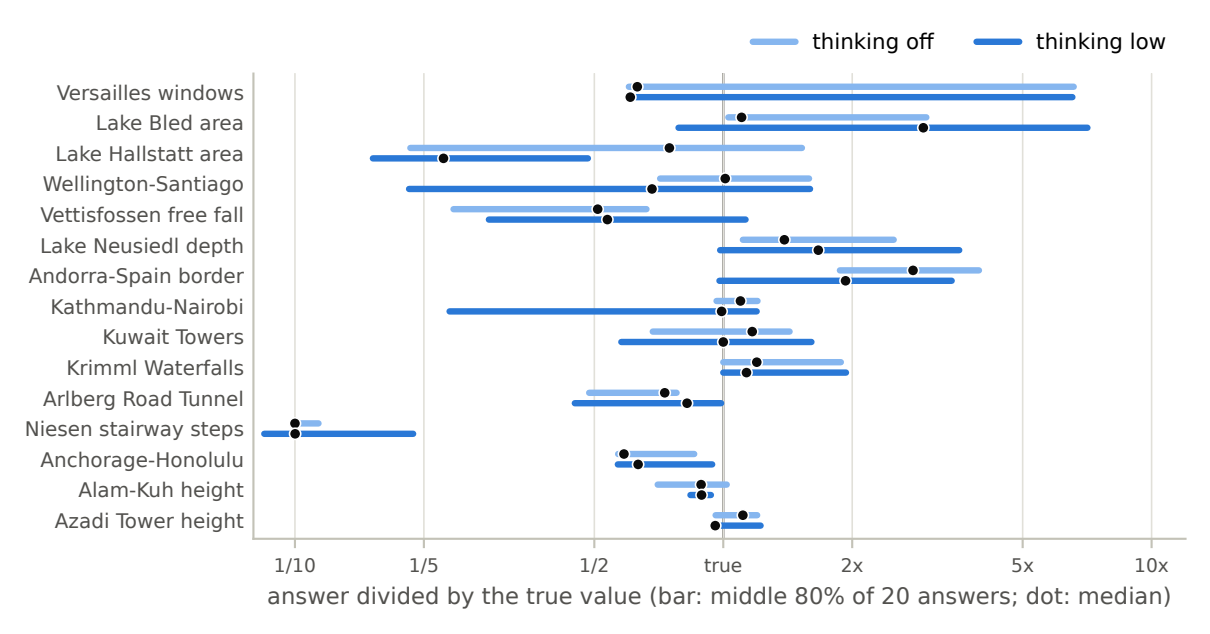

stayed in own normal zone
copied the anchor
landed elsewhere
no number
off, near
29%
52%
10%
9%
off, far
37%
38%
16%
9%
low, near
55%
28%
17%
low, far
55%
18%
26%
high, near
52%
31%
17%
high, far
56%
16%
28%
0
20
40
60
80
100
share of answers (200 per row): thinking level, anchor distance (anchors above the zone)

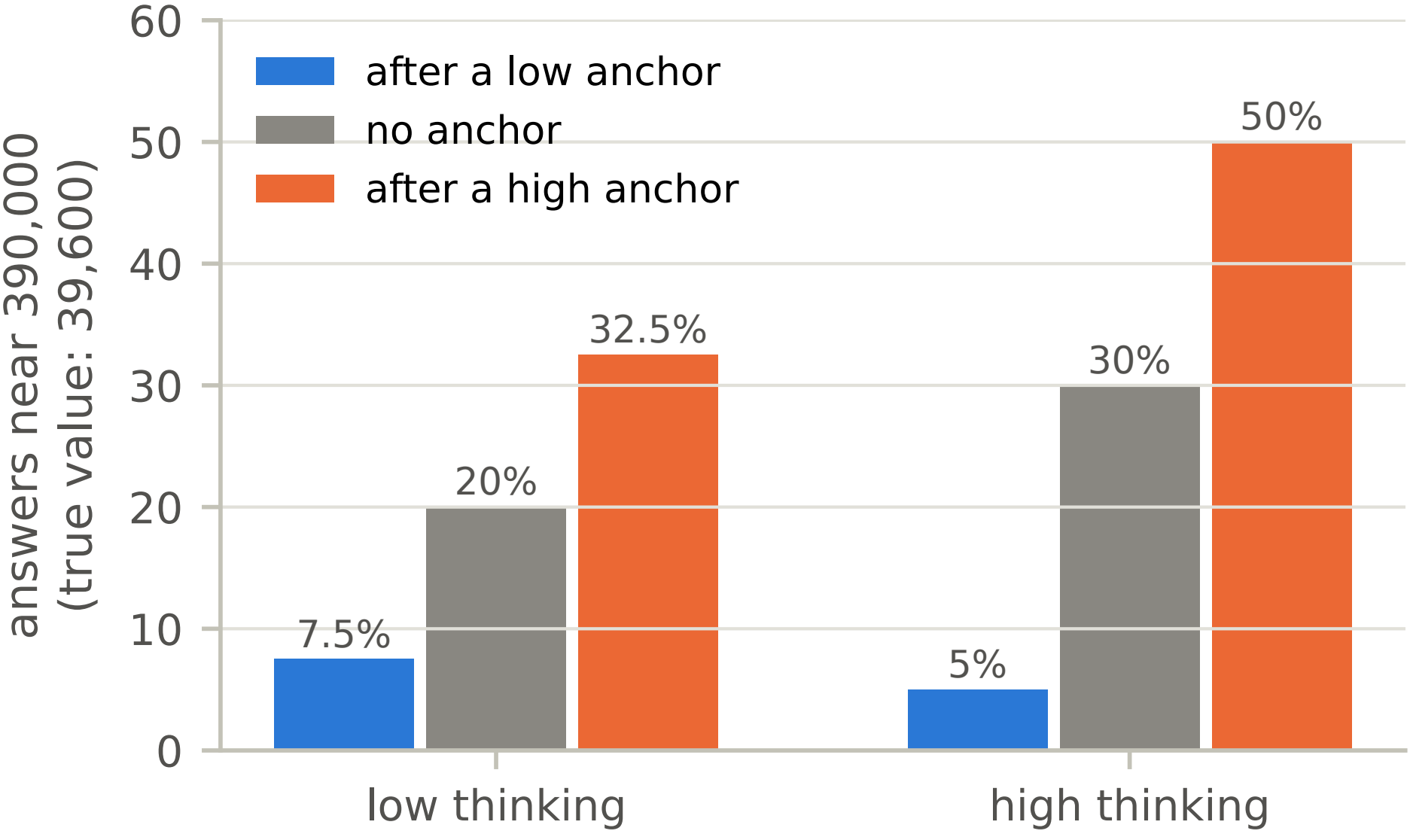

recalled its own value and kept it
23
"remembered" the anchor itself and kept it
8
own value, then pushed towards the anchor
4
other (unit slips, borrowed digits)
4
started at the anchor and adjusted away
1
0
5
10
15
20
25
number of traces (40 read)

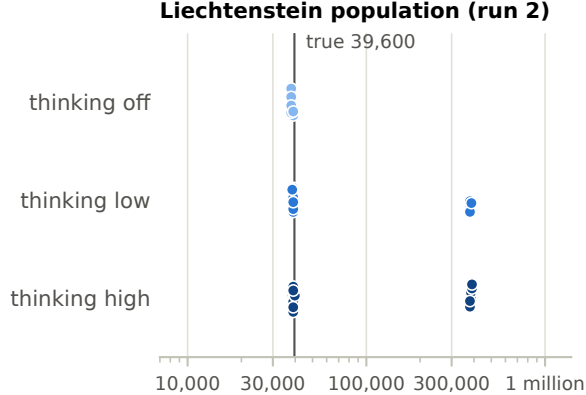

Study 1 questions
Study 2 questions
run 1 (1.5B)
run 2 (32B)
thinking tokens per answer (mean)
0
500
1000
1500
2000
budget 512
budget 2048
431
471
554
671
475
501
946
1569
low
high

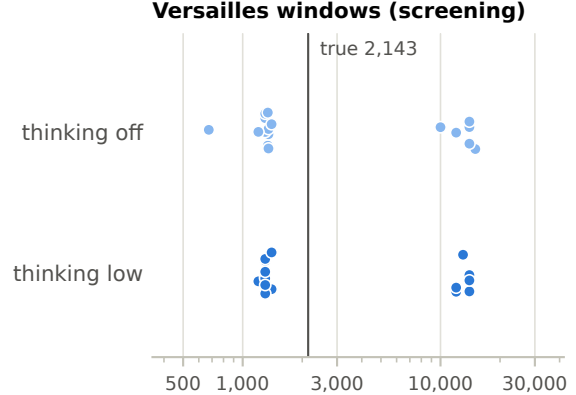


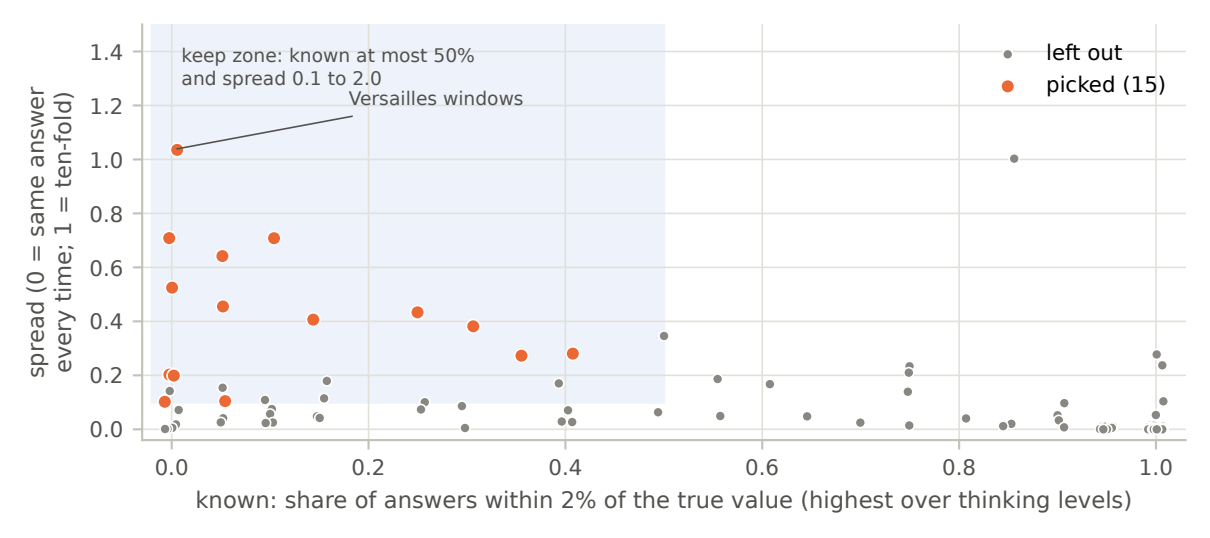

answer divided by the true value (bar: middle 80% of 60 answers without an anchor; dot: median)

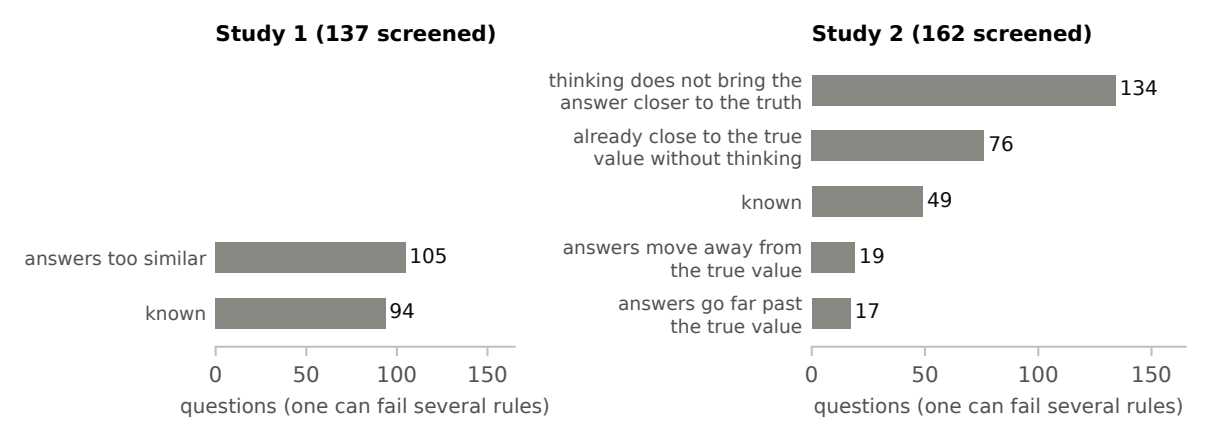

thinking off
thinking low
Versailles windows
Lake Bled area
Lake Hallstatt area
Wellington-Santiago
Vettisfossen free fall
Lake Neusiedl depth
Andorra-Spain border
Kathmandu-Nairobi
Kuwait Towers
Krimml Waterfalls
Arlberg Road Tunnel
Niesen stairway steps
Anchorage-Honolulu
Alam-Kuh height
Azadi Tower height
1/10
1/5
1/2
true
2x
5x
10x
answer divided by the true value (bar: middle 80% of 20 answers; dot: median)

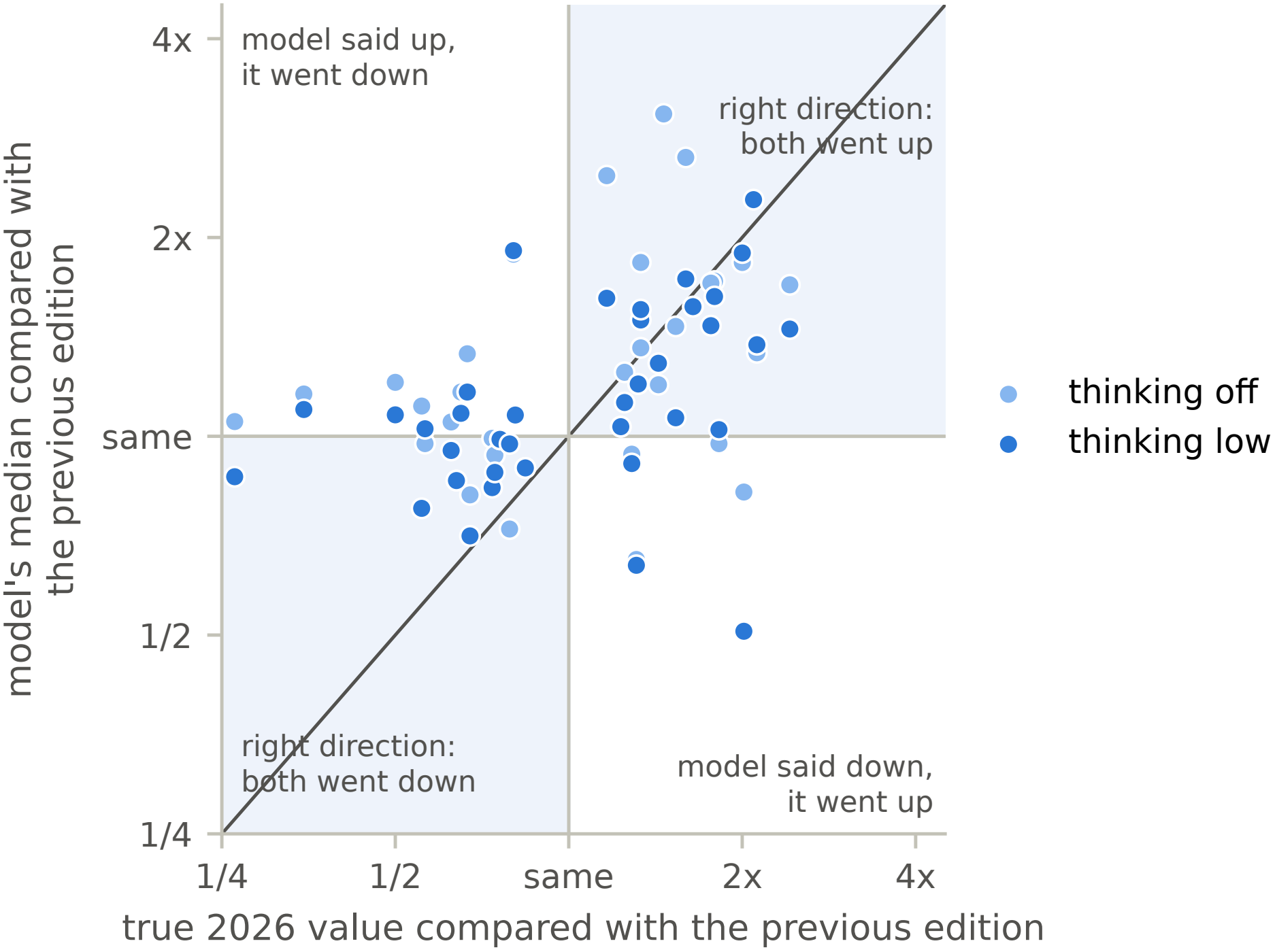

60
50
40
30
20
10
0
answers near 390,000
(true value: 39,600)
after a low anchor
no anchor
after a high anchor
7.5%
20%
32.5%
5%
30%
50%
low thinking
high thinking

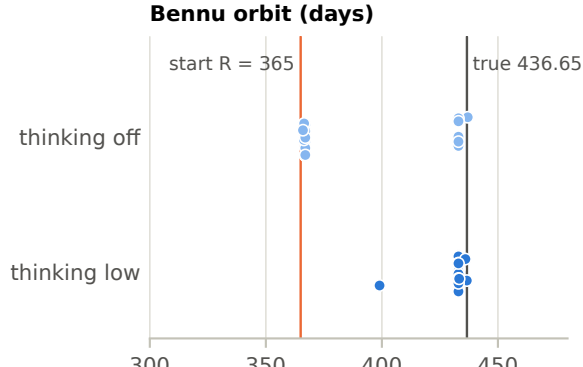

Liechtenstein population (run 2)
true 39,600
thinking off
thinking low
thinking high
10,000
30,000
100,000
300,000
1 million

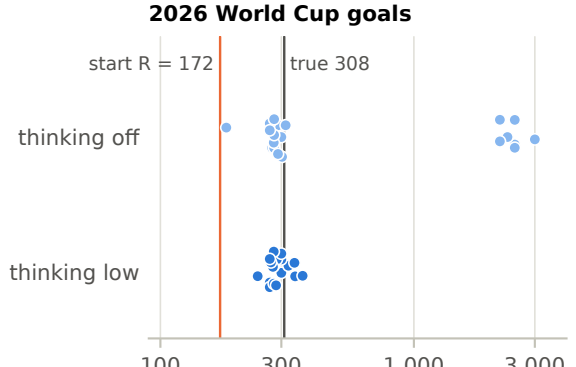

Versailles windows (screening)
true 2,143
thinking off
thinking low
500
1,000
3,000
10,000
30,000

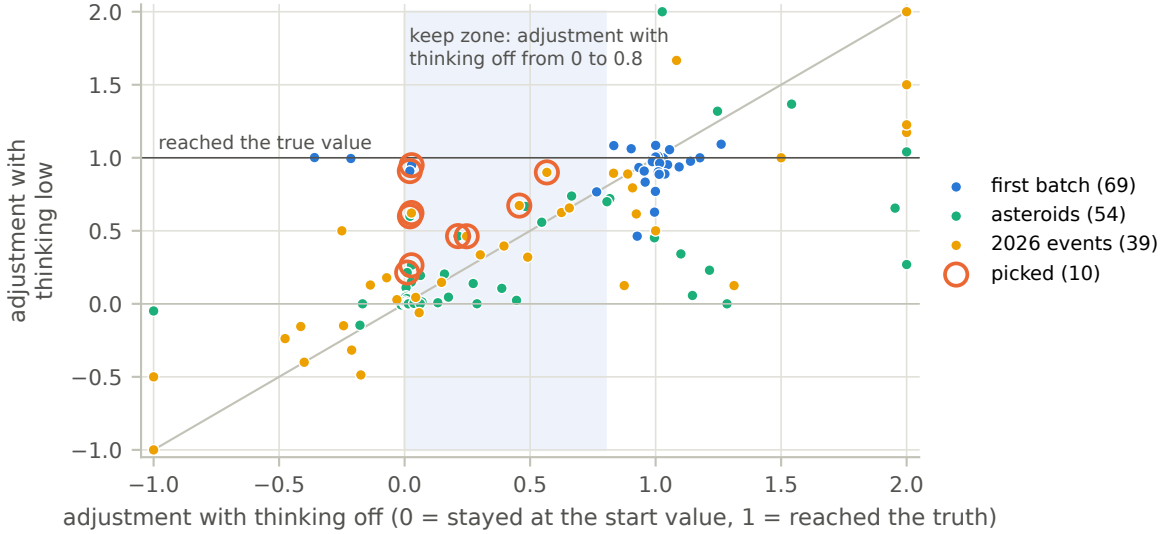

keep zone: known at most 50%
and spread 0.1 to 2.0
Versailles windows
left out
picked (15)
spread (0 = same answer every time; 1 = ten-fold)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
known: share of answers within 2% of the true value (highest over thinking levels)

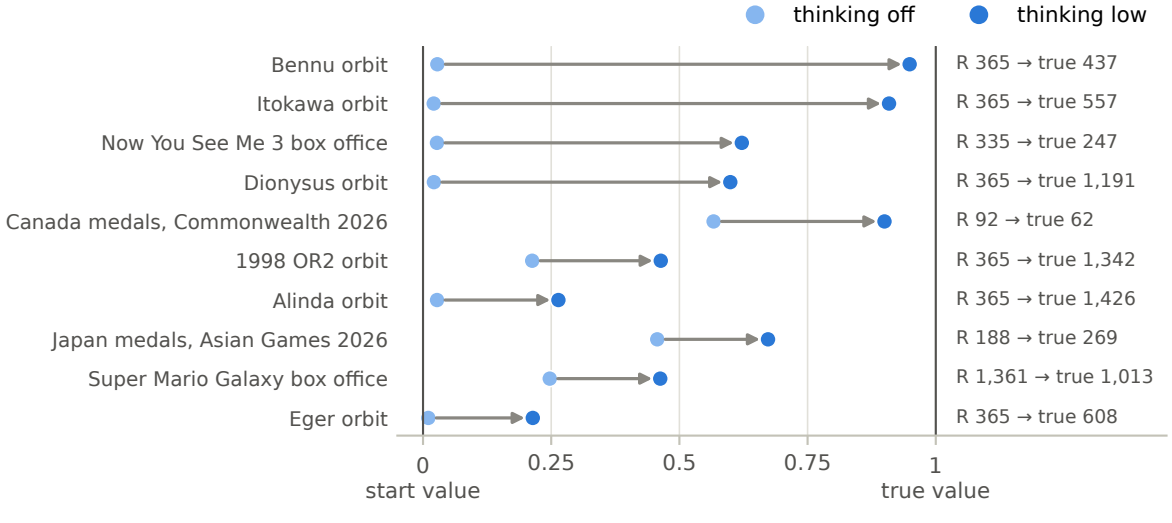

Study 1 (137 screened)
answers too similar
105
known
94
0
50
100
150
questions (one can fail several rules)
Study 2 (162 screened)
thinking does not bring the answer closer to the truth
134
already close to the true value without thinking
76
known
49
answers move away from the true value
19
answers go far past the true value
17
0
50
100
150
questions (one can fail several rules)


model said up,
it went down
right direction:
both went up
right direction:
both went down
model said down,
it went up
4x
2x
same
1/2
1/4
1/4
1/2
same
2x
4x
model's median compared with
the previous edition
true 2026 value compared with the previous edition
thinking off
thinking low


Bennu orbit (days)
start R = 365
true 436.65
thinking off
thinking low
300
350
400
450


2026 World Cup goals
start R = 172
true 308
thinking off
thinking low
100
300
1,000
3,000


keep zone: adjustment with thinking off from 0 to 0.8
reached the true value
adjustment with thinking low
adjustment with thinking off (0 = stayed at the start value, 1 = reached the truth)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
first batch (69)
asteroids (54)
2026 events (39)
picked (10)


thinking off
thinking low
Bennu orbit
Itokawa orbit
Now You See Me 3 box office
Dionysus orbit
Canada medals, Commonwealth 2026
1998 OR2 orbit
Alinda orbit
Japan medals, Asian Games 2026
Super Mario Galaxy box office
Eger orbit
R 365 → true 437
R 365 → true 557
R 335 → true 247
R 365 → true 1,191
R 92 → true 62
R 365 → true 1,342
R 365 → true 1,426
R 188 → true 269
R 1,361 → true 1,013
R 365 → true 608
0
0.25
0.5
0.75
1
start value
true value